

Revenue Analysis

	quarter_year text	cqsales text	pqsales text	qoq text	qoqgrowth numeric
1	Q2-2020	5,542,698.29	7,107,324.5	-1,564,626.21	-22.01
2	Q1-2020	7,107,324.5	9,300,609.76	-2,193,285.26	-23.58
3	Q4-2019	9,300,609.76	11,189,115.52	-1,888,505.76	-16.88
4	Q3-2019	11,189,115.52	6,741,320.18	4,447,795.34	65.98
5	Q2-2019	6,741,320.18	5,268,960.12	1,472,360.06	27.94

Top 3 product lines (Touring Bikes, Mountain Bikes and Road Bikes) were the main drivers for the QoQ decline. Sales dropped due to

1. Lower average selling price (ASP) for Road Bikes, Mountain Bikes and lower quantity sold
2. Slightly higher ASP for Touring Bikes offset by, much lower quantity sold

	quarter_year text	subcategory text	cqsales text	pqsales text	qqsales text	cqqtysold text	pqqtysold text	cqasp numeric	pqasp numeric
1	Q2-2020	Touring Bikes	1,530,842.70	2,328,979.60	-798,136.90	1,707	2,609	896.80	892.67
2	Q2-2020	Mountain Bikes	1,322,358.49	1,815,501.52	-493,143.03	1,524	1,968	867.69	922.51
3	Q2-2020	Road Bikes	1,610,209.29	2,099,650.02	-489,440.73	1,922	2,483	837.78	845.61
4	Q2-2020	Road Frames	175,758.12	76,777.77	98,980.35	445	258	394.96	297.59
5	Q2-2020	Touring Frames	246,807.30	181,738.10	65,069.20	552	458	447.11	396.81
6	Q2-2020	Mountain Frames	343,022.83	309,888.46	33,134.37	979	959	350.38	323.14
7	Q2-2020	Cranksets	23,294.29	35,460.08	-12,165.79	131	172	177.82	206.16

Further analysis of the top 3 contributors of the QoQ variance by region showed the following

1. The US market is the main contributor for the flux for all 3 subcategories
2. For Mountain Bikes in the US market, this is mainly due to lower ASP and lower quantity sold.
3. For Road and Touring Bikes in the US market, it is mainly due to a drop in sales volume offset by slightly higher ASP.

	quarter_year text	subcategory text	country text	cqsales text	pqsales text	qqsales text	cqqtysold text	pqqtysold text	cqasp numeric	pqasp numeric
1	Q2-2020	Mountain Bikes	United States	867,396.17	1,228,519.15	-361,122.98	1,053	1,335	823.74	920.24
2	Q2-2020	Mountain Bikes	Canada	237,495.08	312,548.20	-75,053.12	243	350	977.35	892.99
3	Q2-2020	Mountain Bikes	United Kingdom	61,358.47	115,498.92	-54,140.45	71	138	864.20	836.95
4	Q2-2020	Mountain Bikes	France	133,992.93	105,561.26	28,431.67	132	94	1015.10	1122.99
5	Q2-2020	Mountain Bikes	Germany	22,115.84	39,559.09	-17,443.25	25	41	884.63	964.86
6	Q2-2020	Road Bikes	United States	1,144,154.06	1,440,520.88	-296,366.82	1,334	1,715	857.69	839.95
7	Q2-2020	Road Bikes	United Kingdom	69,755.37	151,710.82	-81,955.45	87	192	801.79	790.16
8	Q2-2020	Road Bikes	Canada	250,673.78	323,301.41	-72,627.63	327	353	766.59	915.87
9	Q2-2020	Road Bikes	France	108,650.78	131,185.43	-22,534.65	126	165	862.31	795.06
10	Q2-2020	Road Bikes	Germany	36,975.30	51,465.47	-14,490.17	48	57	770.32	902.90
11	Q2-2020	Touring Bikes	United States	642,874.65	1,100,908.23	-458,033.58	708	1,226	908.02	897.97
12	Q2-2020	Touring Bikes	Australia	152,702.27	292,480.80	-139,778.53	156	324	978.86	902.72
13	Q2-2020	Touring Bikes	Canada	116,172.91	230,442.83	-114,269.92	134	278	866.96	828.93

Analysing by channel showed that Value Added Reseller (VAR) is the main contributor across all 3 subcategories

For Mountain Bikes and Road Bikes, similar findings as above.

For Touring Bikes, the fall in sales is due to lower ASP and lower sales volume from VAR channel coupled with lower sales volume from Specialty Bike Shop (SBS) channel, slightly offset by higher ASP.

	quarter_year text	subcategory text	country text	businesstype text	cqsales text	pqsales text	qqsales text	cqqtysold text	pqqtysold text	cqasp numeric	pqasp numeric
1	Q2-2020	Mountain Bikes	United States	Value Added Reseller	581,794.94	859,564.38	-277,769.44	751	1,042	774.69	824.92
2	Q2-2020	Mountain Bikes	United States	Warehouse	270,454.34	318,006.14	-47,551.80	291	256	929.40	1242.21
3	Q2-2020	Mountain Bikes	United States	Specialty Bike Shop	15,146.89	50,948.63	-35,801.74	11	37	1376.99	1376.99
4	Q2-2020	Road Bikes	United States	Value Added Reseller	732,157.40	880,503.37	-148,345.97	814	999	899.46	881.38
5	Q2-2020	Road Bikes	United States	Specialty Bike Shop	111,316.81	189,901.16	-78,584.35	149	262	747.09	724.81
6	Q2-2020	Road Bikes	United States	Warehouse	300,679.85	370,116.35	-69,436.50	371	454	810.46	815.23
7	Q2-2020	Touring Bikes	United States	Value Added Reseller	171,642.10	486,080.78	-314,438.68	200	561	858.21	866.45
8	Q2-2020	Touring Bikes	United States	Specialty Bike Shop	41,053.53	142,258.22	-101,204.69	40	148	1026.34	961.20
9	Q2-2020	Touring Bikes	United States	Warehouse	430,179.02	472,569.23	-42,390.21	468	517	919.19	914.06

Gross Profit Analysis

Although the current quarter's revenue has dropped, the gross profits have increased due to higher QoQ gross margins.

	quarter_year text	cqgp text	pqgp text	qoqgp text	cqgpm numeric	pqgpm numeric
1	Q2-2020	447,251.07	435,492.22	11,758.85	8.07	6.13
2	Q1-2020	435,492.22	753,686.79	-318,194.57	6.13	8.10

Analysing by subcategory showed that the increase in gross profit (GP) of \$11.8k mainly driven by

- Lower volume of Touring Bikes sold but at less negative margins (\$32.7k)
- Higher volume of products such as Road Frames, Pedals, Mountain Frames, Handlebars and Helmets sold at similar / lower margins (\$15.5k)

Offset by,

- Lower volume of Mountain bikes sold offset by higher margins (-\$30.3k)
- Lower volume of Cranksets and Shorts sold (-\$6.0k)

	quarter_year text	subcategory text	cqgp text	pqgp text	qoqgp text	cqgpm numeric	pqgpm numeric	cqqtysold text	pqqtysold text
1	Q2-2020	Touring Bikes	-8,022.15	-40,693.30	32,671.15	-0.52	-1.75	1,707	2,609
2	Q2-2020	Mountain Bikes	352,498.49	382,797.77	-30,299.28	26.66	21.08	1,524	1,968
3	Q2-2020	Road Frames	19,608.22	11,729.83	7,878.39	11.16	15.28	445	258
4	Q2-2020	Cranksets	6,056.03	10,119.00	-4,062.97	26.00	28.54	131	172
5	Q2-2020	Touring Frames	19,957.99	23,942.31	-3,984.32	8.09	13.17	552	458
6	Q2-2020	Pedals	6,202.34	4,102.08	2,100.26	25.93	25.99	645	410
7	Q2-2020	Shorts	15,724.33	17,728.17	-2,003.84	36.31	36.41	1,088	1,183
8	Q2-2020	Mountain Frames	29,955.47	27,961.58	1,993.89	8.73	9.02	979	959
9	Q2-2020	Handlebars	3,440.90	1,649.16	1,791.74	26.00	25.99	279	191
10	Q2-2020	Helmets	8,081.13	6,370.21	1,710.92	36.72	37.65	1,064	806

Based on the data above, Mountain bikes contributed to the bulk of the decline in GP which is offset by an increase in GP from Touring Bikes and Road Frames.

The lower QoQ GP of -\$30.3k from Mountain bikes were mainly contributed by

- Lower sales volume in United States, offset by higher margins contributed -\$42.0k
- Lower sales volume and lower margins in United Kingdom contributed -\$19.1k

Offset by,

- Higher sales volume and higher margins in France of \$26.2k
- Higher margins in Canada, offset by lower sales volume of \$4.9k

	quarter_year text	subcategory text	country text	qoqgp text	cqgpm numeric	pqgpm numeric	cqqtysold text	pqqtysold text
1	Q2-2020	Mountain Bikes	United States	-41,996	27.03	22.50	1,053	1,335
2	Q2-2020	Mountain Bikes	France	26,171	33.74	18.03	132	94
3	Q2-2020	Mountain Bikes	United Kingdo...	-19,159	16.67	25.44	71	138
4	Q2-2020	Mountain Bikes	Canada	4,871	24.42	16.99	243	350
5	Q2-2020	Mountain Bikes	Australia	-1,254	0.00	9.08	0	10
6	Q2-2020	Mountain Bikes	Germany	1,067	21.06	9.08	25	41

Analyzing further into the Mountain Bikes' sales channels for the top 3 contributing countries showed the following.

United States

	quarter_year text	businesstype text	subcategory text	country text	cqsales text	cqcost text	cqgp text	pqsales text	pqcost text	pqgp text	cqgpm numeric	pqgpm numeric	cqunitprice numeric	pqunitprice numeric	cqqtysold numeric	pqqtysold numeric
7	Q2-2020	Value Added Resel...	Mountain Bik...	United States	581,794.94	414,565.29	167,229.65	859,564.38	668,390.36	191,174.02	28.74	22.24	774.69	824.92	751	1042
8	Q2-2020	Specialty Bike Shop	Mountain Bik...	United States	15,146.89	13,771.78	1,375.11	50,948.63	41,373.27	9,575.36	9.08	18.79	1376.99	1376.99	11	37
9	Q2-2020	Warehouse	Mountain Bik...	United States	270,454.34	204,635.49	65,818.85	318,006.14	242,336.38	75,669.76	24.34	23.80	929.40	1242.21	291	256

- Warehouse (WH) : Although quantity sold increased after a reduction in price, GP was affected negatively despite slightly higher Gross Profit Margins (GPM).
- VAR: Decrease in price did not increase the sales volume which caused GP to drop significantly despite higher GPM.
- SBS: The drop in GP was mainly contributed by lower GPM.

United Kingdom

	quarter_year text	business_type text	subcategory text	country text	qsales text	qcost text	qgpp text	psales text	pqcost text	ppgpp text	qgppm numeric	pgppm numeric	qunitprice numeric	punitprice numeric	qqty numeric	pqty numeric
4	Q2-2020	Specialty Bike Shop	Mountain Bik...	United Kingdo...	2,753.98	2,503.96	250.02	2,753.98	2,503.96	250.02	9.08	9.08	1376.99	1376.99	2	2
5	Q2-2020	Warehouse	Mountain Bik...	United Kingdo...	22,079.78	20,075.33	2,004.45	34,538.72	31,403.16	3,135.56	9.08	9.08	1003.63	1233.53	22	28
6	Q2-2020	Value Added Resel...	Mountain Bik...	United Kingdo...	36,524.71	28,552.72	7,971.99	78,206.22	52,206.59	25,999.63	21.83	33.24	777.12	724.13	47	108

- WH: Decrease in price did not increase the sales volume which caused GP to drop.
- VAR: GP has dropped mainly from lower GPM. Increase in price has also resulted in lower sales volume.
- SBS: No changes.

France

	quarter_year text	business_type text	subcategory text	country text	qsales text	qcost text	qgpp text	psales text	pqcost text	ppgpp text	qgppm numeric	pgppm numeric	qunitprice numeric	punitprice numeric	qqty numeric	pqty numeric
1	Q2-2020	Value Added Resel...	Mountain Bik...	France	20,016.61	11,899.48	8,117.13	25,092.88	22,814.92	2,277.96	40.55	9.08	690.23	784.15	29	32
2	Q2-2020	Warehouse	Mountain Bik...	France	104,337.39	68,125.79	36,211.60	70,829.45	54,949.28	15,880.17	34.71	22.42	1086.85	1287.81	96	55
3	Q2-2020	Specialty Bike Shop	Mountain Bik...	France	9,638.93	8,763.86	875.07	9,638.93	8,763.86	875.07	9.08	9.08	1376.99	1376.99	7	7

- WH: Increase in GP mainly due to higher GPM. Decrease in price has also resulted in higher sales volume, further increasing GP.
- VAR: Increase in GP mainly due to higher GPM offset by lower prices as it did not result in higher sales volume.
- SBS: No changes.

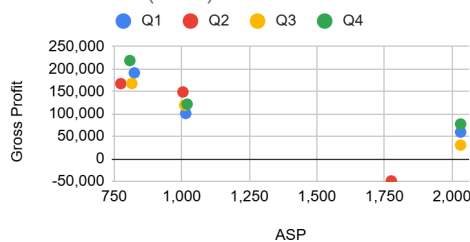
Insights

Based on the latest Q2 results, it seems that lowering prices showed limited effectiveness in increasing sales volume for Mountain Bikes. The Warehouse channel in France was the only channel where a reduction in price clearly resulted in a higher quantity sold and higher GP in Q2 (excluding effects from higher GPM).

The WH channel in the United States had a drop in GP even though a price reduction led to higher sales volume which showed that lowering prices to increase sales volume may not always result in higher profitability.

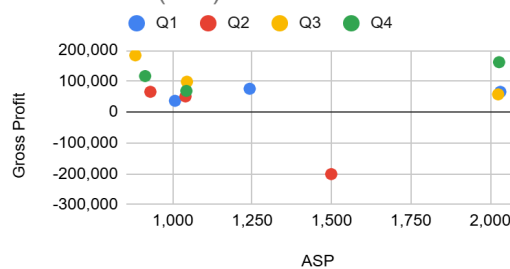
Further analysis of the US market for Mountain Bikes seemed to suggest that a reduction in prices (ASP in the range of \$750 - \$1,000) worked best for the VAR channel as it consistently increases gross profits.

ASP vs GP (VAR)

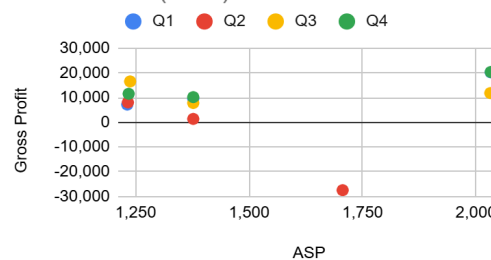


For SBS and WH channel, it showed mixed results where gross profits are higher either when ASP is set at lower price range (WH: around \$750 - \$1,250, SBS: around \$1,200 - \$1,375) or when ASP is high (WH and SBS: around \$2,050).

ASP vs GP (WH)

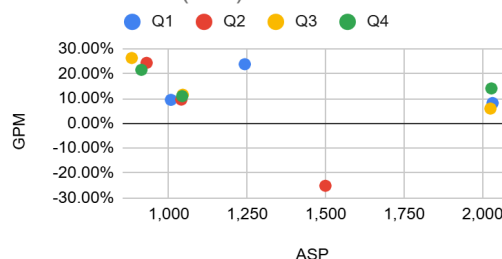


ASP vs GP (SBS)

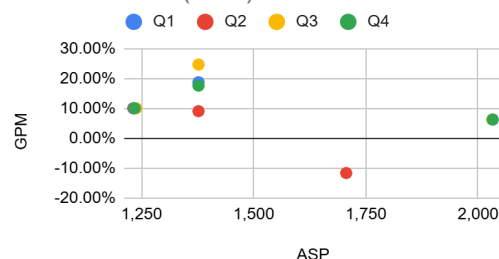


Comparing ASP vs GPM for WH and SBS channels showed that the GPM are typically higher at lower ASP which suggest that lowering ASP to drive sales volume may increase GPM possibly via economies of scale.

ASP vs GPM (WH)

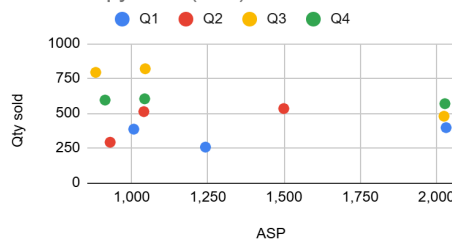


ASP vs GPM (SBS)

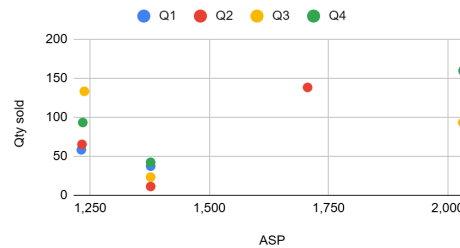


However, plotting ASP against quantity sold for SBS channel showed lower ASP typically resulted in lower quantities sold. For WH channel, the scatter plot seemed to suggest that demand is more price sensitive in Q3. In Q1, Q2 and Q4, prices and demand seemed less correlated.

ASP vs qty sold (WH)



ASP vs qty sold (SBS)

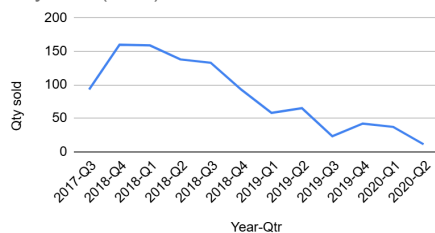


Looking at the ASP & quantity sold over time.

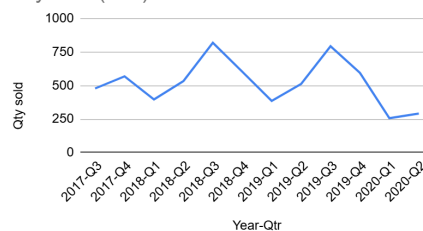
For WH channel, there seems to be a seasonality effect at play and demand/sales volume is usually higher in Q3. Lower prices seemed to also spur higher sales volume in Q3.

For SBS, demand/sales volume has been falling consecutively despite a general trend of lower prices over time.

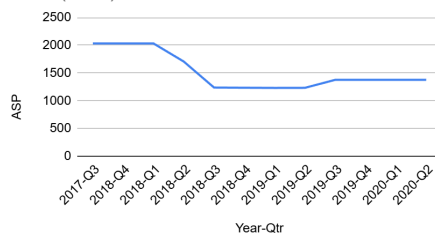
Qty sold (SBS)



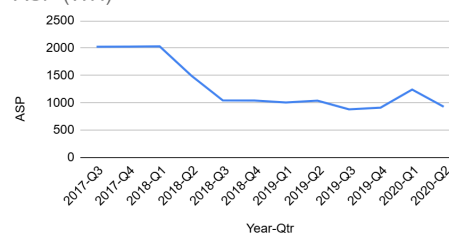
Qty sold (WH)



ASP (SBS)



ASP (WH)



Conclusion & Recommendations

Based on the findings above, I would recommend the following for Mountain Bikes in the US market.

For the VAR channel, we should price Mountain Bikes in the range of \$750 - \$1,000 as data points have shown that lower prices would increase gross profit.

For the WH channel, Mountain Bikes should be priced at a range of around \$800 - \$1,100 in Q3 to drive higher sales volume and gross profit. For the rest of the year, reduction in prices is not recommended due to weak correlation in prices and demand to maintain profitability.

For the SBS channel, given that lowering prices do not drive sales volume, we should price in the range of \$1,250 - \$1,375 to maintain profitability. Additionally, given the small scale of the SBS channel and declining sales despite reduction in prices in the US market, management should also consider if we should exit this market and reallocate the resources to other regions/channels.

Limitations

Given the limited amount of data points, some of these data points may be skewed by one-time occurrence effects. More data points are required to better understand the impact on seasonality demand if any.

More information on the cost structure for each channel is also required to analyse the GPM trends better and isolate impact of ASP on gross profits

More information on competitors' pricing strategy required for a more effective pricing strategy.

Follow-up actions

1. Work with sales/pricing team to better understand pricing strategy, seasonality demand and cost structure of the different channels to better analyse GPM
2. Analyse variances contributed by other product lines/channels/regions and evaluate their pricing strategies
3. Assess if there are any channel/country that we should also consider exiting

Dataset Source: AdventureWorks

<https://www.kaggle.com/datasets/algorismus/adventure-works-in-excel-tables?resource=download>